

Composite Design Pattern (scaffold)

This module is a **placeholder** — the pattern is not implemented here yet. The full, runnable, production-mirrored example lives in `Design-Pattern-RealUseCase/RealUseCase_Structural/RealUseCase_Composite_Design_Pattern`.

Target UML (what the implementation should look like)

```
<<abstract>> Component
| +operation() |
| ^           ^
| |           |
Leaf      Composite
|           |
|           | -children : List<Component>
|           | +operation() -> for each child.operation()
```

Current code, line by line

```
package com.design.patterns.composite;

public class CompositeDesignPattern {

    public static void main(String[] args) {
        System.out.println("Composite Design Pattern");
    }
}
```

- `package com.design.patterns.composite;` — the pattern's dedicated package, ready for the real classes.
- `public class CompositeDesignPattern` — the module's entry class, named after the pattern.
- `public static void main(String[] args)` — standard JVM entry point so the scaffold runs standalone.
- `System.out.println("Composite Design Pattern");` — prints the pattern name, proving the module builds and runs; replace with a real demo when implementing.